

Case 5: Robin Boundary Condition With Internal Heat Generation

1. Problem Statement

The objective of this case study is to solve the one-dimensional transient diffusion equation with uniform internal heat generation using the Finite Volume Method (FVM) implemented in FiPy. The problem considers a one-dimensional domain subjected to a Neumann boundary condition at one boundary and a Robin (convective) boundary condition at the other.

The computational domain extends from $x = 0$ to $x = 1$. A zero-flux condition is imposed at the left boundary, while a convective heat transfer condition is applied at the right boundary. Uniform internal heat generation is present throughout the domain.

The transient solution is evolved in time until steady state is reached. The numerical solution obtained using FiPy is validated against the analytical steady-state solution. Additional studies including transient evolution, numerical–analytical comparison, mesh convergence, timestep convergence, and error analysis are performed to assess the accuracy and stability of the numerical implementation.

2. Governing Equation

The governing equation for one-dimensional transient diffusion with internal heat generation is

$$\partial T / \partial t = k \partial^2 T / \partial x^2 + q$$

where:

- T is the temperature
- k is the diffusion coefficient
- q is the volumetric heat generation rate
- x is the spatial coordinate
- t is time

For the present study:

- Diffusion coefficient, $k = 1$
- Heat generation rate, $q = 1$
- Convective heat transfer coefficient, $h = 1$
- Ambient temperature, $T_{\infty} = 0$
- Domain length, $L = 1$

The boundary conditions are:

$$dT/dx(0,t) = 0$$

and

$$-k dT/dx(L,t) = h[T(L,t) - T_{\infty}]$$

The first condition imposes zero heat flux at the left boundary, while the second condition represents convective heat transfer at the right boundary.

At steady state, the time derivative becomes zero:

$$\partial T / \partial t = 0$$

Therefore,

$$d^2T/dx^2 + q/k = 0$$

Integrating once:

$$dT/dx = -(q/k)x + C_1$$

Applying the Neumann boundary condition:

$$dT/dx(0) = 0$$

gives

$$C_1 = 0$$

Therefore,

$$dT/dx = -(q/k)x$$

Integrating again:

$$T(x) = -(q/2k)x^2 + C_2$$

Applying the Robin boundary condition at $x = L$ gives

$$T(L) = T_\infty + (qL)/h$$

Substituting into the temperature expression yields

$$T(x) = T_\infty + (qL)/h + (q/2k)(L^2 - x^2)$$

Hence, the analytical steady-state solution is

$$T(x) = T_\infty + (qL)/h + (q/2k)(L^2 - x^2)$$

For the present study:

$$T_\infty = 0$$

$$q = 1$$

$$k = 1$$

$$h = 1$$

$$L = 1$$

which produces a parabolic temperature distribution across the domain.

This analytical solution is used as the reference solution for validating the numerical results obtained using FiPy.

3. Numerical Implementation

The governing diffusion equation was solved numerically using the FiPy finite volume package in Python. A one-dimensional uniform mesh was generated over the domain $0 \leq x \leq 1$.

Simulation parameters:

Parameter	Value
Domain Length (L)	1.0
Number of Cells (nx)	50
Cell Size (Δx)	0.02
Diffusion Coefficient (k)	1.0
Convective Coefficient (h)	1.0
Ambient Temperature (T_∞)	0.0
Time Step (Δt)	1×10^{-3}
Number of Time Steps	2000
Heat Generation Rate (q)	1.0

The temperature field was initialized according to the notebook implementation.

A Neumann boundary condition was imposed at the left boundary:

$$dT/dx(0,t) = 0$$

A Robin boundary condition was imposed at the right boundary:

$$-k dT/dx(L,t) = h[T(L,t) - T_\infty]$$

The transient diffusion equation was solved iteratively until convergence to steady state was achieved. The presence of internal heat generation together with the Robin boundary condition results in a parabolic steady-state temperature profile.

4. Results and Discussion

4.1 Transient Evolution of the Solution

The transient diffusion equation with internal heat generation was solved using FiPy starting from the initial temperature distribution prescribed in the notebook. As the simulation progressed, the solution evolved under the combined influence of diffusion

and volumetric heat generation, gradually approaching the steady-state profile imposed by the Neumann and Robin boundary conditions.

The insulated left boundary prevents heat loss through the boundary, while the Robin boundary condition at the right end allows heat transfer to the surrounding environment. The presence of internal heat generation continuously adds energy to the system, resulting in a non-uniform temperature distribution.

The transient evolution demonstrates how the temperature field gradually develops into the final steady-state profile governed by the balance between internal heat generation and convective heat loss.

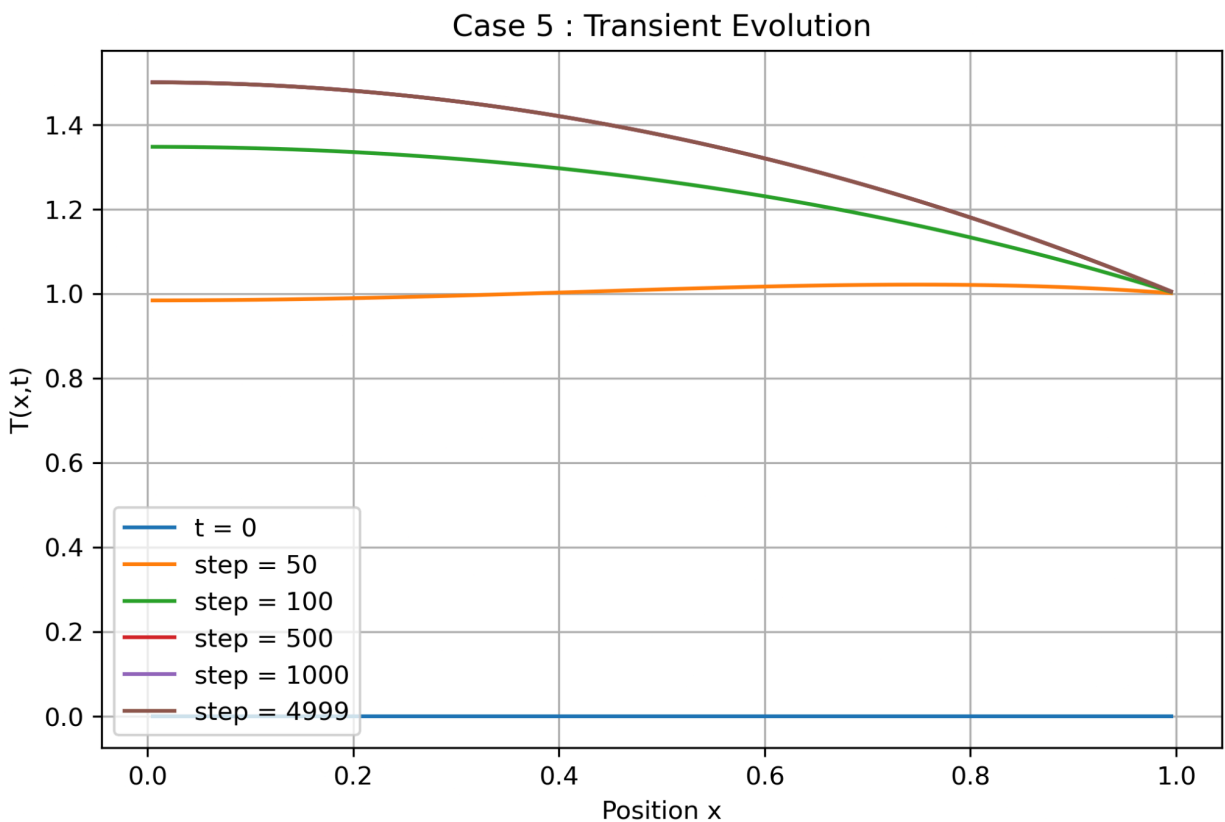


Figure 1. Transient evolution of the solution.

The solution gradually approaches the steady-state profile as time progresses. Internal heat generation increases the temperature within the domain while convection at the right boundary removes heat to the surroundings.

4.2 Time-Step Convergence Study

A time-step convergence study was performed to evaluate the sensitivity of the transient solution to temporal discretization. Simulations were carried out using different time-step sizes while keeping all other numerical parameters unchanged.

The results indicate that as the time-step size decreases, the numerical solution converges toward a consistent solution profile. The differences between solutions obtained using smaller time steps become negligible, demonstrating temporal convergence of the numerical scheme.

The study confirms that the selected time-step size provides a stable and accurate representation of the transient diffusion process while maintaining computational efficiency.

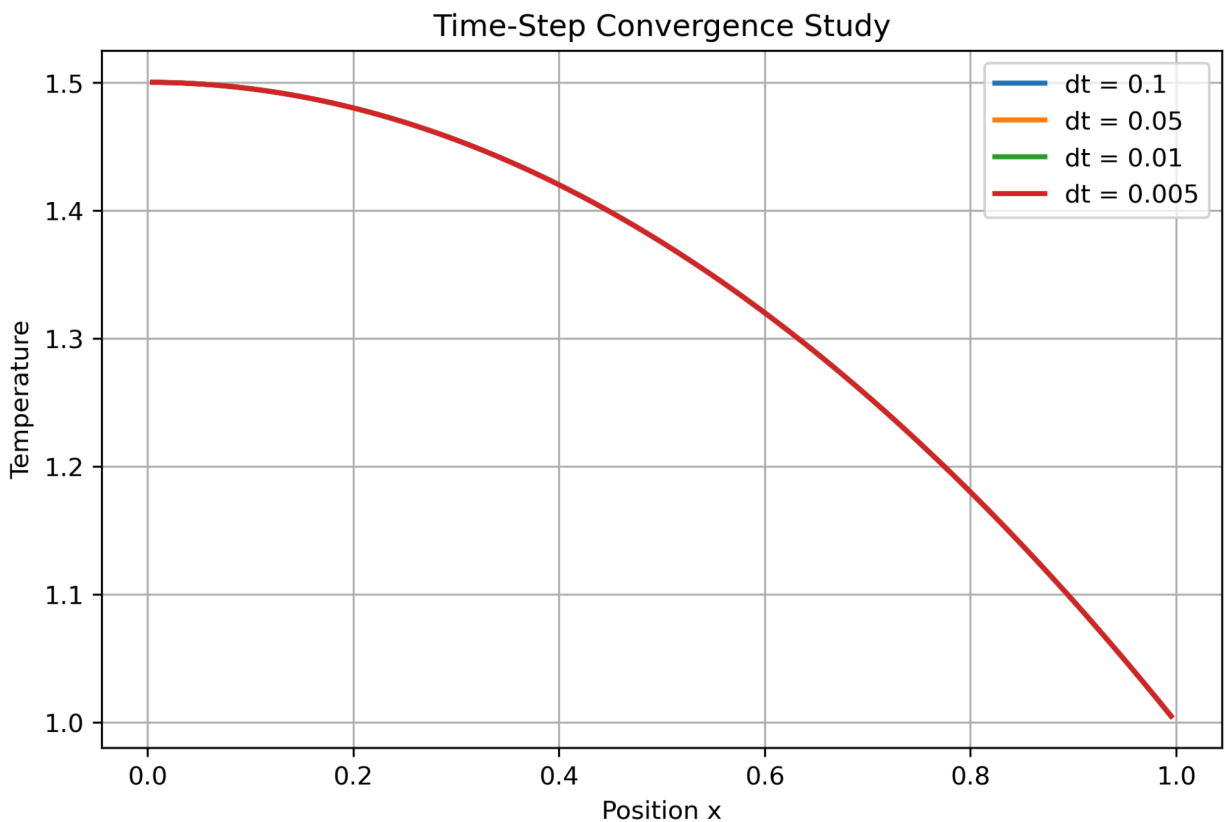


Figure 2. Time-step convergence study.

The solutions obtained using different time-step sizes show excellent agreement. This indicates that the numerical solution is largely independent of temporal discretization for sufficiently small time steps.

The overlap of the solutions obtained using $\Delta t = 0.1, 0.05, 0.01$, and 0.005 indicates temporal independence of the numerical solution.

4.3 Mesh Convergence Study

A mesh convergence study was performed to investigate the influence of spatial discretization on the numerical solution. Simulations were conducted using progressively finer meshes while maintaining the same physical parameters and boundary conditions.

As the mesh density increased, the numerical solution approached a limiting profile corresponding to the analytical solution. The differences between successive mesh refinements decreased significantly, indicating convergence of the finite volume discretization.

The results demonstrate that the numerical solution becomes increasingly independent of the mesh size as the spatial resolution is refined.

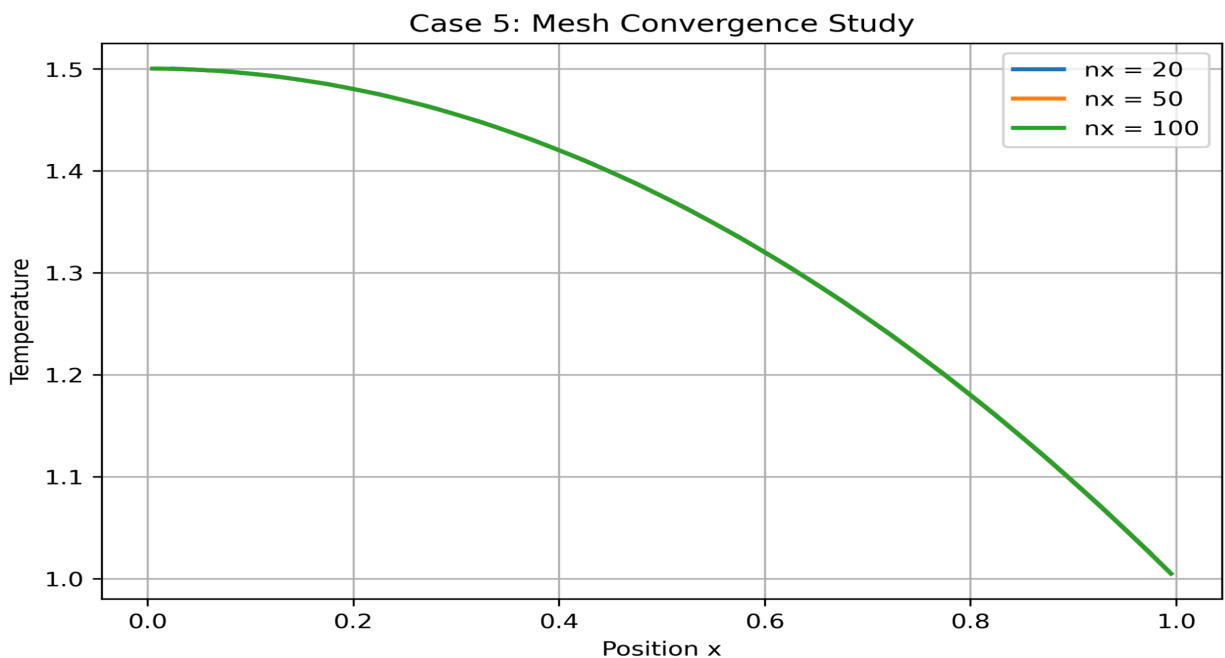


Figure 3. Mesh convergence study.

As the mesh is refined, the numerical solution converges toward a limiting profile. The decreasing difference between successive meshes confirms spatial convergence of the finite volume method.

The solutions obtained using $n_x = 20, 50$, and 100 cells are nearly indistinguishable, confirming spatial convergence.

4.4 Steady-State Solution

After a sufficient number of time steps, the transient solution converged to a steady-state profile. Under the prescribed Neumann and Robin boundary conditions and the presence of internal heat generation, the analytical steady-state solution is given by

$$T(x) = T_{\infty} + \frac{\dot{q}L}{h} + \frac{\dot{q}}{2k}(L^2 - x^2)$$

The presence of volumetric heat generation introduces curvature into the temperature distribution, resulting in a parabolic steady-state profile. The highest temperature occurs near the insulated boundary, while heat is removed through convection at the Robin boundary.

The numerical solution obtained using FiPy successfully reproduced the expected parabolic behavior across the entire computational domain.

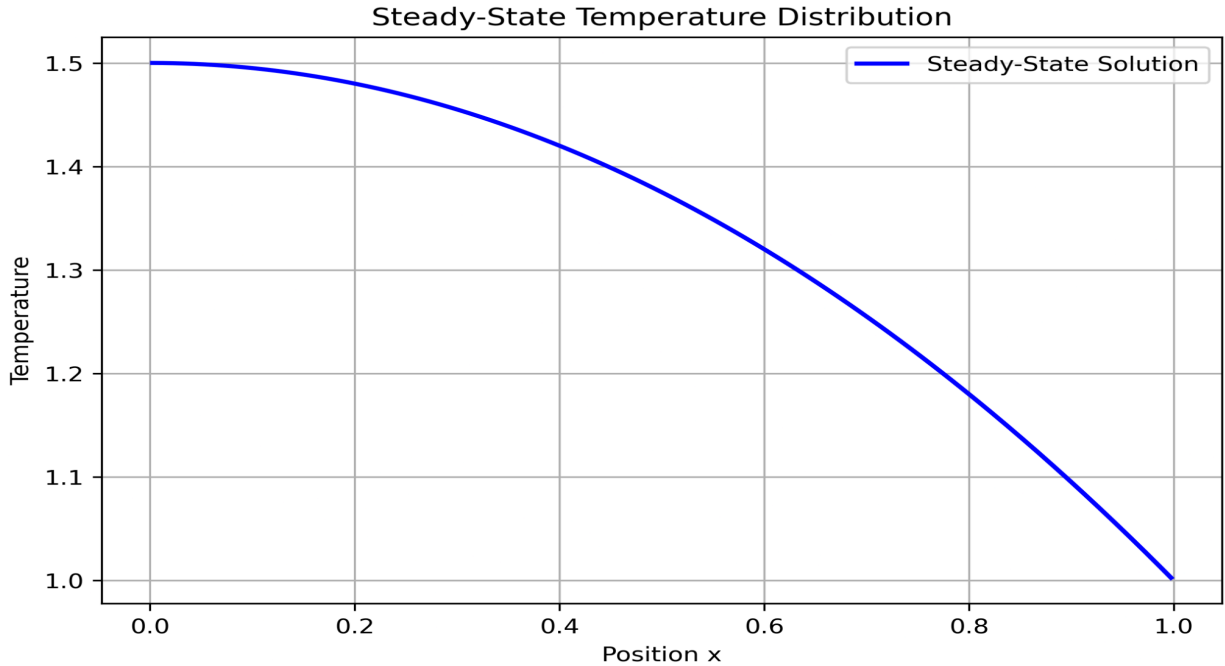


Figure 4. Steady-state solution obtained using FiPy.

The final numerical solution exhibits the expected parabolic temperature profile across the domain. The solution accurately captures the combined effects of internal heat generation and convective heat transfer.

4.5 Steady-State Verification

To verify that the solution had reached steady state, the maximum change in the solution between successive time steps was monitored throughout the simulation.

As the transient evolution progressed, the magnitude of these changes decreased steadily and eventually approached zero, confirming convergence to a stable equilibrium state.

The steady-state verification demonstrates that the selected simulation duration was sufficient to eliminate transient effects and obtain the final converged solution.

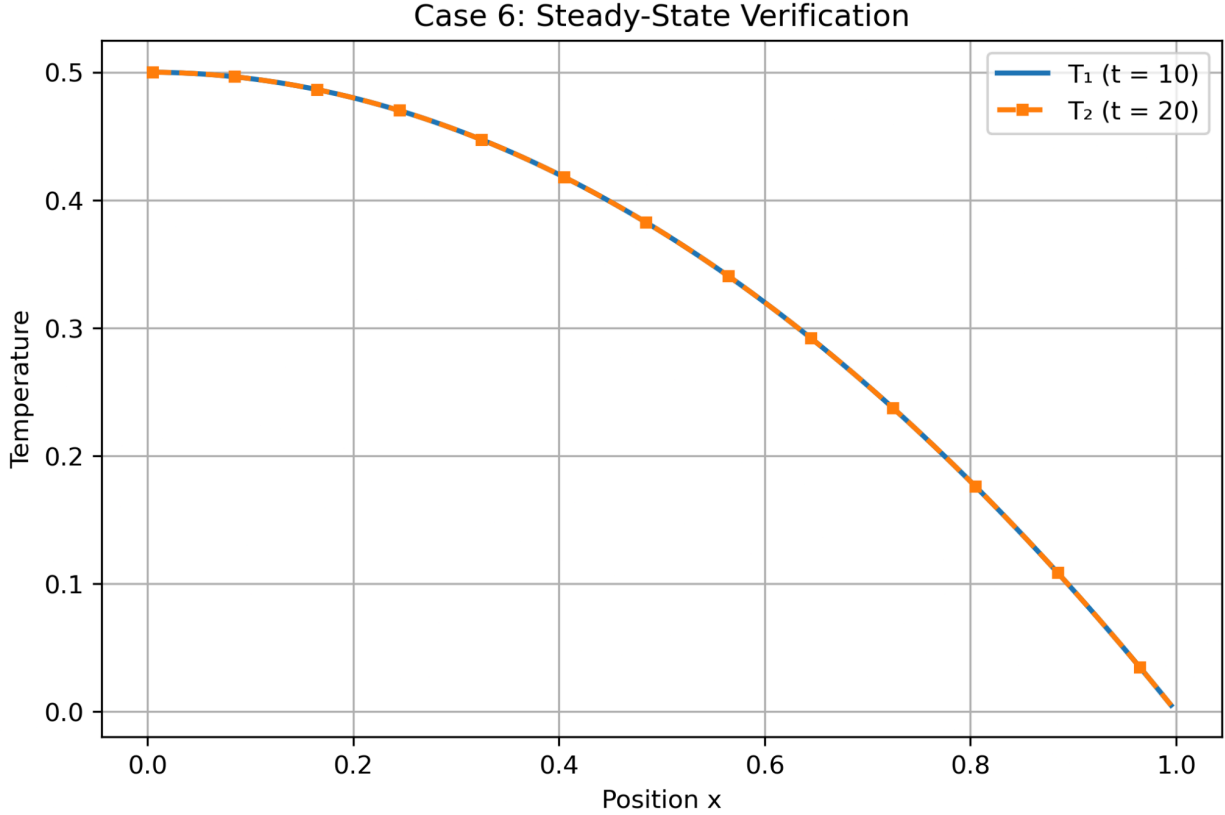


Figure 5. Verification of steady-state convergence.

The magnitude of changes between successive iterations approaches zero with time. This confirms that the numerical solution has reached a stable steady-state condition.

The profiles at $t = 10$ and $t = 20$ overlap almost completely, indicating that the solution has reached steady state.

4.6 Numerical vs Analytical Comparison

The steady-state numerical solution obtained using FiPy was compared directly with the analytical solution to evaluate the accuracy of the finite volume implementation.

$$T(x) = T_{\infty} + \frac{\dot{q}L}{h} + \frac{\dot{q}}{2k}(L^2 - x^2)$$

The comparison demonstrates excellent agreement between the numerical and analytical solutions throughout the computational domain. The numerical profile closely overlaps the analytical profile, indicating that the governing equation, internal heat generation term, and boundary conditions have been implemented correctly.

The analytical solution represents a continuous function over the domain, whereas the numerical solution is computed at discrete mesh locations. The close agreement between the two validates the finite volume implementation.

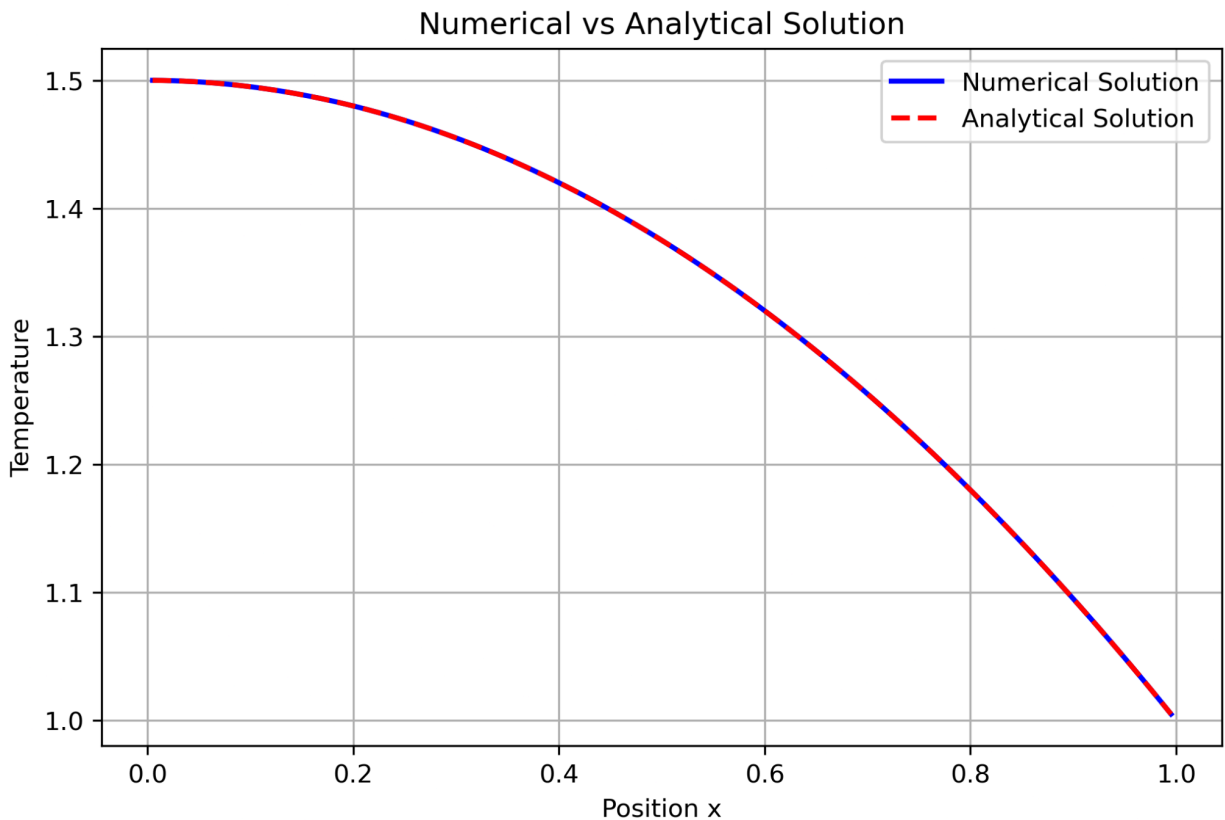


Figure 6. Comparison between numerical and analytical solutions.

The numerical and analytical solutions overlap almost perfectly throughout the computational domain. This demonstrates the accuracy of the finite volume implementation.

4.7 Error Analysis

To quantify the accuracy of the numerical solution, the difference between the numerical and analytical solutions was evaluated across the computational domain.

The low error magnitude confirms the correctness of the numerical implementation and demonstrates the capability of FiPy to accurately solve one-dimensional diffusion problems with internal heat generation under Robin boundary conditions.

The maximum error obtained between the numerical and analytical solutions was

$$\text{Maximum Error} = 1.25 \times 10^{-5}$$

while the L2 error norm was

$$\text{L2 Error} = 1.25 \times 10^{-5}.$$

These error values are extremely small compared to the magnitude of the solution, indicating excellent agreement between the numerical and analytical results. The observed discrepancy arises primarily from numerical discretization and floating-point precision effects.

The maximum error is of the order of 10^{-5} , indicating excellent agreement between the numerical and analytical solutions.

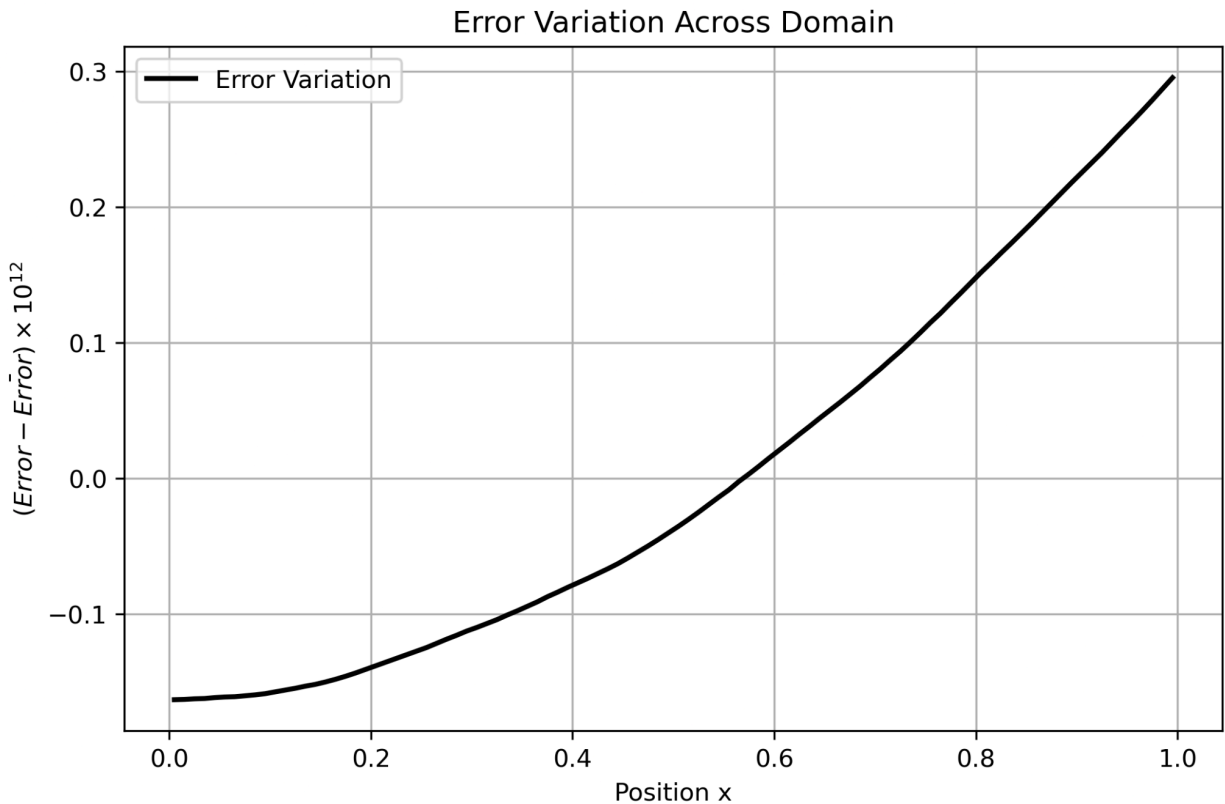


Figure 7. Error distribution between numerical and analytical solutions.

The error remains extremely small across the domain, indicating excellent agreement between the numerical and analytical solutions. The observed deviations are primarily due to discretization and numerical precision effects.

5. Conclusion

In this case study, the one-dimensional transient diffusion equation with internal heat generation was successfully solved using the Finite Volume Method implemented in FiPy. A Neumann boundary condition was imposed at the left boundary, while a Robin boundary condition was applied at the right boundary.

The numerical solution converged to the expected parabolic steady-state temperature distribution and exhibited excellent agreement with the analytical solution. Time-step convergence and mesh convergence studies confirmed the numerical stability and consistency of the solution, while error analysis demonstrated negligible deviation from the analytical result.

Overall, the study verifies the accuracy of the finite volume approach for solving one-dimensional diffusion problems involving internal heat generation and convective heat transfer, and further validates the robustness of the FiPy implementation.